

Chapter 14: More About Chances

Chap 13:

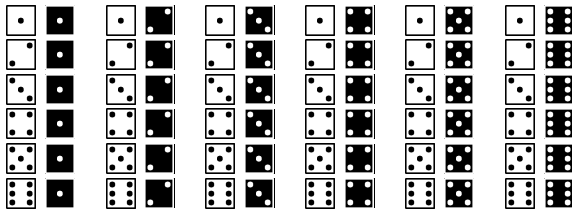
- ▶ Complementation Rule
- ▶ Conditional Chance
- ▶ Multiplication Rule
- ▶ Independence & Independence Rule

Chap 14:

- ▶ Addition Rule

A pair of dice are to be thrown. What is the chance of getting a total of 7 spots?

- ▶ There are 36 possible ways for 2 dice to fall:



- ▶ By symmetry, all these ways are equally likely, so each one has chance $1/36$ to happen.
- ▶ There are 6 ways to get a total of 7 spots:

- ▶ The chance of getting a total of 7 spots equals $6/36$.

When figuring chances, one helpful strategy is to write down a complete list of all the possible ways that the chance process can turn out.

The Addition Rule

For two events A and B , will the chance that A happens or B happens equals the sum of the chance of the individual chances?

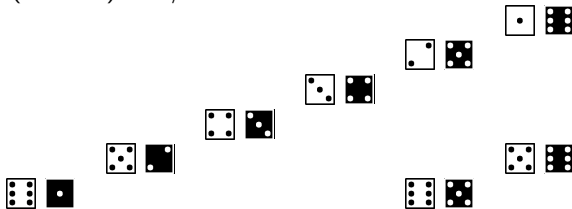
$$P(A \text{ or } B)? = P(A) + P(B)$$

Let us look at some examples

Example 1

A pair of dice are to be thrown. What is the chance of getting a total of 7 spots, or of 11 spots?

- ▶ Event A: The total is 7 spots. $P(A) = \frac{6}{36}$.
- ▶ Event B: The total is 11 spots. $P(B) = \frac{2}{36}$.
- ▶ There are 8 ways to get a total of 7 or 11 spots.
 $P(A \text{ or } B) = 8/36$.



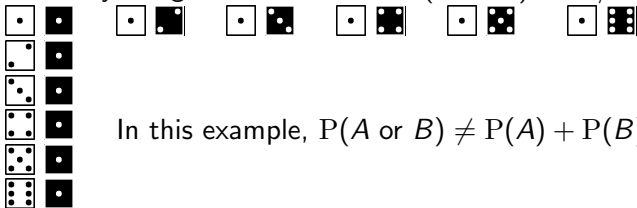
- ▶ In this example, $P(A \text{ or } B) = P(A) + P(B)$

Observe the two events A and B are **mutually exclusive** of each other. You cannot get a total of 7 spots at the same time as getting a total of 11 spots.

Example 2

Two dice — one white, one black — are to be thrown. What is the chance of getting at least one ace?

- ▶ Event A: The white die lands ace. $P(A) = \frac{6}{36}$.
- ▶ Event B: The black die lands ace. $P(B) = \frac{6}{36}$.
- ▶ Getting at least 1 ace means A happens or B happens. There are 11 ways to get at least 1 ace. $P(A \text{ or } B) = 11/36$.



- ▶ Q: Why does blindly adding the individual chances give the wrong answer?
- ▶ A: Adding the individual chances double-counts the chance that both dice land aces:

When are Two Things Mutually Exclusive?

- ▶ Two things are mutually exclusive when the occurrence of one prevents the occurrence of the other.
- ▶ Suppose a white die and a black die are to be thrown. The following outcomes are mutually exclusive:
 - ▶ Getting a total of 7.
 - ▶ Getting a total of 11.
- ▶ The following outcomes are NOT **mutually exclusive**:
 - ▶ Getting an ace on the white die.
 - ▶ Getting an ace on the black die.

The Addition Rule

- ▶ If two things are **mutually exclusive**, then the chance that at least one of those things will happen equals the sum of the individual chances.

$$P(A \text{ or } B) = P(A) + P(B).$$

- ▶ If the two things are NOT mutually exclusive, adding the individual chances will give the wrong answer, due to double-counting.

$$P(A \text{ or } B) < P(A) + P(B)$$

The Addition Rule for Three or More Things

Q : Is there an addition rule for three or more mutually exclusive events?

A : Yes.

The chance that at least one of several things will happen equals the sum of the individual chances, provided that the occurrence of any one of the things prevents the occurrence of each of the other ones.

$$P(A \text{ or } B \text{ or } C \text{ or } \dots) = P(A) + P(B) + P(C) + \dots$$

“Independent” versus “Mutually Exclusive”

- ▶ Two things are **independent**
if the conditional chances for the 2nd one given the 1st are the same, no matter how the first one turns out.

$$P(B|A) = P(B|\text{not } A) = P(B)$$

- ▶ Two things are **mutually exclusive**
when the occurrence of one prevents the occurrence of the other.

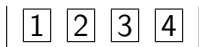
$$P(A \text{ and } B) = 0$$

They are different concepts.

- ▶ If A and B are independent, $P(A \text{ and } B) = P(A) \times P(B)$.
- ▶ If A and B are mutually exclusive, $P(A \text{ and } B) = 0$.

“Independent” versus “Mutually Exclusive”: Example 1

Two tickets will be drawn at random *with replacement* from the box



Consider these two events:

- ▶ Event A: The 1st ticket drawn is

4

.
- ▶ Event B: The 2nd ticket drawn is

4

.

True or false:

- ▶ True Events A and B are independent.
- ▶ False Events A and B are mutually exclusive.

“Independent” versus “Mutually Exclusive”: Example 2

As in Example 1, but for drawing *WITHOUT* replacement

True or false

- ▶ False Events A and B are independent.
 - ▶ Given that A happens, the conditional chance for B equals 0.
 - ▶ Given that A doesn't happen, the conditional chance for B equals 1/3.
- ▶ True Events A and B are mutually exclusive.

Conclusion

Whether or not two things are independent, or mutually exclusive, depends not only on the things themselves, but also on the chance process involved!

When to Add, and When to Multiply?

- ▶ Try to visualize the chance process that the problem is about — throwing dice, dealing cards, or whatever it is.
- ▶ Identify the event whose chance is asked for.
- ▶ If this event is complicated, try to break it into simpler things whose chances you know. For example,
 - ▶ $C = (A \text{ and } B)$
 C is the event that two simpler events A and B both happen.
 - ▶ $C = (A \text{ or } B)$
 C is the event that at least one of the two simpler events A and B happens.

When to Add, and When to Multiply? (Cont'd)

If $C = (A \text{ and } B)$, think whether A and B are **independent**.

- ▶ If yes, use the **independence rule**:

$$P(A \text{ and } B) = P(A) \times P(B).$$

- ▶ If no, use the **multiplication rule**:

$$P(A \text{ and } B) = P(A) \times P(B|A).$$

If $C = (A \text{ or } B)$, think whether A and B are **mutually exclusive**.

- ▶ If yes, use the **addition rule**:

$$P(A \text{ or } B) = P(A) + P(B) \text{ .}$$

- ▶ If no, then

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B).$$

(This formula is beyond the scope of STAT20000)

When to Add, and When to Multiply? (Cont'd)

Exercise 1: A die is rolled 6 times;

- (a) The chance that the first roll is an ace and the last roll is an ace is (ii)
- (b) The chance that the first roll is an ace or the last roll is an ace is (iii)

Options:

(i) $\frac{1}{6} + \frac{1}{6}$ (ii) $\frac{1}{6} \times \frac{1}{6}$ (iii) neither of these

Remark: The chance in (b) is

$$P(A) + P(B) - P(A \text{ and } B) = \frac{1}{6} + \frac{1}{6} - \frac{1}{36}$$

When to Add, and When to Multiply? (Cont'd)

Exercise 2: A deck of cards is shuffled.

- (a) The chance that the top card is the ace of ♠ and the bottom card is an ace of ♠ is (iii)
- (b) The chance that the top card is the ace of ♠ or the bottom card is an ace of ♠ is (i)

Options:

$$(i) \frac{1}{52} + \frac{1}{52} \quad (ii) \frac{1}{52} \times \frac{1}{52} \quad (iii) \text{ neither of these}$$

Remark:

The chance in (a) is 0.

The ace of ♠ cannot turn up in two places.

When to Add, and When to Multiply? (Cont'd)

True or false

If you see the words “mutually exclusive,” add the chances.

If you see the word “independent,” multiply the chances.

This is False . You first have to think about what chance you need to find.

Solving a complicated problem may involve several steps,
some using the addition rule,
some using the multiplication rule,
and some using the complementation rule.

Full House (Revisited)

In poker, what's the chance of being dealt a full house (one pair and one triple)?

- Think about where the two cards that give you the pair lie among the 5 cards you're dealt. The possibilities are:

Event	Card Order				
A_1	P	P	T	T	T
A_2	P	T	P	T	T
A_3	P	T	T	P	T
A_4	P	T	T	T	P
A_5	T	P	P	T	T
A_6	T	P	T	P	T
A_7	T	P	T	T	P
A_8	T	T	P	P	T
A_9	T	T	P	T	P
A_{10}	T	T	T	P	P

Full House (Revisited)

Event	Card Order	► Are $A_1, \dots, A_{10}$ mutually exclusive? <u>Yes</u>
A_1	P P T T T	► So $P(\text{full house}) = P(A_1) + \dots + P(A_{10})$.
A_2	P T P T T	
A_3	P T T P T	► $P(A_1) = \frac{52}{52} \times \frac{3}{51} \times \frac{48}{50} \times \frac{3}{49} \times \frac{2}{48} = 0.0144\%$.
A_4	P T T T P	
A_5	T P P T T	► $P(A_2) = \frac{52}{52} \times \frac{48}{51} \times \frac{3}{50} \times \frac{3}{49} \times \frac{2}{48} = 0.0144\%$.
A_6	T P T P T	
A_7	T P T T P	► $\vdots$
A_8	T T P P T	
A_9	T T P T P	► $A_3, A_4, \dots, A_{10}$ all have the same chance 0.0144%.
A_{10}	T T T P P	
		► So $P(\text{full house}) = 10 \times 0.0144\% = 0.144\%$.

This problem was solved by first listing the ways, then using the multiplication rule to find the chance of each way, and finally using the addition rule to add the chances.

Summary

- ▶ When figuring chances, one helpful strategy is to write down a complete list of all the possible ways that the chance process can turn out. If this is too hard, at least write down a few typical ways, and count how many ways there are in total.
- ▶ The chances that at least one of several things will happen equals the sum of the individual chances, provided that the things are mutually exclusive. Otherwise, adding the chances will give the wrong answer, due to double counting.
- ▶ “Independent” and “mutually exclusive” are not synonyms.
- ▶ If you are having trouble working out the chance of an outcome, try to figure out the chance of its opposite; then subtract from 100%.
- ▶ Solving a complicated problem may involve several steps, some using the addition rule, some using the multiplication rule, and some using the complementation rule.